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$$PPPM = \mathbf{W}^\top \cdot \mathbf{I} = (\sigma_1 \ \sigma_2 \ \sigma_3 \ \sigma_4) \cdot (MW \ IWW \ AAP \ GT)^\top. \quad (1)$$

$$\mathbf{I} = \mathbf{T}(\cdot) = \begin{pmatrix} \alpha(t) \\ \beta(t) \\ \theta(t) \\ \gamma(t) \end{pmatrix} = \begin{pmatrix} -0.0481 \times t + 97.3493 \\ 0.0503 \times t - 100.6650 \\ 0.0355 \times t - 70.7723 \\ 0.0386 \times t - 76.9758 \end{pmatrix} \quad (2)$$

since we applied the MLP as the backbone model, weight matrix have to be calculated as :

$$\begin{cases} \mathbf{O} = \mu(\mathbf{W}_1 \cdot \mathbf{X} + \mathbf{b}) \\ \hat{\mathbf{Y}} = \mathbf{W}_2 \cdot \mathbf{O} + \mathbf{b} \end{cases}, \text{ where } \mu(\cdot) = \text{ReLU}(\cdot) \quad (3)$$

$$\hat{\mathbf{Y}} = \mathbf{W}_2 \cdot \mu(\mathbf{W}_1 \cdot \mathbf{X} + \mathbf{b}) = \mathbf{W}_{\text{total}} \cdot \mathbf{X} + \mathbf{b}_{\text{total}} \quad (4)$$

where:

$$\mathbf{W}_{\text{total}} = \mathbf{W}_2 \cdot \mu(\cdot) \cdot \mathbf{W}_1 = (\varepsilon_1 \ \varepsilon_2) \cdot \begin{pmatrix} \mu(x_i)' & 0 & 0 & 0 \\ 0 & \mu(x_i)' & 0 & 0 \\ 0 & 0 & \mu(x_i)' & 0 \\ 0 & 0 & 0 & \mu(x_i)' \end{pmatrix} \cdot \begin{pmatrix} \omega_{11} & \dots & \omega_{14} \\ \omega_{21} & \dots & \omega_{24} \end{pmatrix} \quad (5)$$

through training we have:

$$\mathbf{W}_2 = (-0.48617 \ 0.47315), \ \mathbf{W}_1 = \begin{pmatrix} 0.31434 & -0.27920 & -0.15085 & -0.18880 \\ -0.31076 & 0.26576 & 0.14595 & 0.18557 \end{pmatrix} \quad (6)$$

Hence,

$$\mathbf{W}^\top = (-0.29985 \ 0.26148 \ 0.14239 \ 0.17959) \approx (-0.30 \ 0.26 \ 0.14 \ 0.18) \quad (7)$$

then we have:

$$PPPM = -0.3 \times MW + 0.26 \times IWW + 0.14 \times AAP + 0.18 \times GT \quad (8)$$

if being mapped into the time dimension, then we have:

$$PPPM = 0.039 \times t - 79.14 \quad (9)$$

where t stands for years and PPPM is a dimensionless variable which reflects the annual profits of construction projects. If $PPPM \geq 0$, then any construction projects can make profits without destroying the environments. Contrarily, if $PPPM < 0$, it may destroy the environments while carrying out the project.

To give this equation stronger practical significance, we introduced "time coefficient" which describes the starting date of a project, which goes as :

$$PPPM = 0.039 \times (t + t_0) - 79.14 \quad (10)$$

From this end, t stands for the lasting time of a specified project. Therefore, the optimized model goes as:

$$PPPM = 0.039 \times (t_0 + \Delta t) - 79.54 \quad (11)$$

Similarly, if $PPPM \geq 0$, it can make profits without destroying the environment.

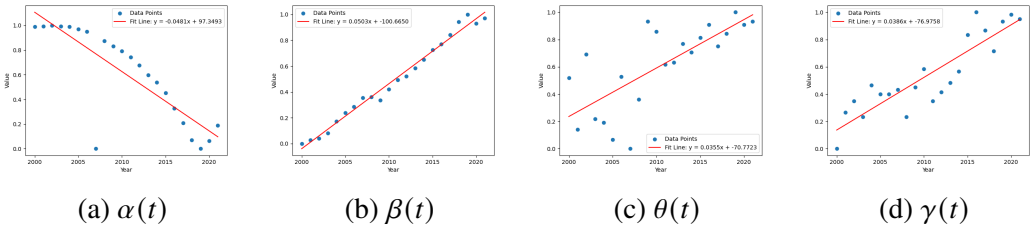


Figure 1: Visualization of the linear fitting process.